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THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6689-6694 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6696-6716, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Reductions of schemes, fibres of a morphism over a generic point, and the interaction of reduction with base change; the comparison is retained.

Source reference. Original lemma algebra-lemma-minimal-prime-reduced-ring: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6716–6716. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-tensor-localization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6716–6716. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $\eta \in Y$ be a generic point of an irreducible component of Y. Then $(X_\eta)_{red} = (X_{red})_\eta$.*

Proof. Choose an affine neighbourhood $\mathrm{Spec}(A) \subset Y$ of η. Choose an affine open $\mathrm{Spec}(B) \subset X$ mapping into $\mathrm{Spec}(A)$ via the morphism f. Let $\mathfrak{p} \subset A$ be the minimal prime corresponding to η. Let B_{red} be the quotient of B by the nilradical $\sqrt{(0)}$. The algebraic content of the lemma is that $C = B_{red} \otimes_A \kappa(\mathfrak{p})$ is reduced. Denote $I \subset A$ the nilradical so that $A_{red} = A/I$. Denote $\mathfrak{p}_{red} = \mathfrak{p}/I$ which is a minimal prime of A_{red} with $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}_{red})$. Since $A \to B_{red}$ and $A \to \kappa(\mathfrak{p})$ both factor through $A \to A_{red}$ we have $C = B_{red} \otimes_{A_{red}} \kappa(\mathfrak{p}_{red})$. Now $\kappa(\mathfrak{p}_{red}) = (A_{red})_{\mathfrak{p}_{red}}$ is a localization by Algebra, Lemma algebra-lemma-minimal-prime-reduced-ring. Hence C is a localization of B_{red} (Algebra, Lemma algebra-lemma-tensor-localization) and hence reduced. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.